

CORRECTED SHOULD-COST ANALYSIS — Bosch IPB2.0 ECU PCBA

Commodity: PCBA (populated assembly) · Currency: GBP · Region: China · Annual volume: 150,000 · Operations: 4

Re-costed from eight board photographs after the uploaded tool report costed this board as bare *pcb_fab* with zero operations. Bottom-up, fully traceable.

TOTAL SHOULD-COST	MATERIAL	PROCESS	LABOUR	OVERHEAD	MARGIN
£184.76	72.6%	10.6%	1.0%	7.6%	7.4%

§1 — Key Assumptions

Board 170×110 mm (187 cm²) · 6 layers · FR4 High-Tg 150C · 2 oz outer / 1 oz inner copper · ENIG · AEC-Q Grade 2 / PPAP
755 SMT placements · 3 BGA · 2 press-fit connectors · double-sided assembly · assembly quality grade AEC-Q Grade 1 (brake-by-wire)
Overhead 9% of factory base · Margin 8% of subtotal · Yield 98.5% first-pass · NRE £46,000 assembly + £9,500 fab, both amortised over 150,000.
Board size and layer count are scaled from the photographs, not measured. A caliper measurement and a layer count from a cross-section would remove the largest single uncertainty in this estimate.

§2 — 8-Bucket Cost Breakdown

Cost Bucket	Amount (GBP)	% of Total
1. Raw Material (bare PCB + BOM + yield)	£134.23	72.6%
2. Process (Machine)	£19.65	10.6%
3. Direct Labour	£1.94	1.0%
4. Tooling (amortised)	£0.31	0.2%
5. Packaging	£0.35	0.2%
6. Logistics	£0.55	0.3%
FACTORY COST	£157.02	85.0%
7. Overhead (SG&A)	£14.05	7.6%
SUBTOTAL	£171.07	92.6%
8. Supplier Margin	£13.69	7.4%
TOTAL SHOULD-COST	£184.76	100.0%

§3 — Bill of Materials (the uploaded tool report contained no BOM at all)

Ref	Qty	Description	Unit £	Ext £	Source
✓ U-MCU	1	Renesas R7F702300B RH850 automotive MCU	15.000	15.00	marking R7F702300B FAB-A-C 2308 JAPAN
✓ U-ASIC1	1	Bosch 40342/01 motor-control ASIC	10.000	10.00	marking 40342 /01 2324 VC252NVN
✓ U-ASIC2	1	Bosch 40341/01 ASIC	7.000	7.00	marking 40341 /01 A2192F4A
✓ U-ASIC3	1	Bosch 2302701 ASIC	4.000	4.00	marking 2302701
✓ U-SBC	1	ST L9369 multi-channel driver / SBC	5.000	5.00	marking L9369 VC DTE
~ U-BGA2	1	Secondary safety MCU / ASIC (BGA)	8.000	8.00	package visible, marking not legible
~ U-MEM	1	External memory (BGA)	3.000	3.00	package visible, marking not legible
✓ Q1-6	6	Q142E power stage / dual half-bridge	3.000	18.00	marking Q142E BABA W31C ×6
✓ U-CAN	2	NXP TJA1463A CAN-FD SIC transceiver	1.200	2.40	marking TJA1463A
✓ U-71H	3	Bosch 71H740 driver / power-path	1.500	4.50	marking 71H740 Prm 2321 C3
✓ U-76E2	1	Bosch 76E240 device	1.800	1.80	marking 76E240 PNW 2328 B6
✓ U-76E8	1	Bosch 76E840 device	1.800	1.80	marking 76E840 PEm 2319 A2
✓ U-7S1R	2	Bosch 7S1R540H power device	1.600	3.20	marking 7S1R540H Prt 2313D6
✓ C-BULK	6	Polymer alu cap 150 µF 35 V (351 150 EJV)	0.400	2.40	marking 351 150 EJV ×6
~ L-PWR	8	Shielded power inductor 0.47–10 µH	0.600	4.80	0.47µH/10µH legible; count from both sides
~ R/C-SMD	620	MLCC + thick-film resistors, AEC-Q200	0.009	5.58	density count across both populated sides
~ D/Q-SML	90	Small-signal semis (SOT/SOD/QFN)	0.090	8.10	density count; individual markings not legible
~ Y1	1	Crystal / oscillator	0.450	0.45	package visible near MCU
~ J1-2	2	Automotive press-fit header (main + motor)	2.500	5.00	through-hole field, bottom-left
✓ TIM	8	Thermal interface pad (power stages)	0.120	0.96	pink pads visible on underside
BOM TOTAL			110.99		

✓ = package marking legible in a photograph (13 of 20 lines, 69% of BOM value). ~ = count or price inferred from board density and class medians. Component prices are China buy, 150k/yr, and carry roughly ±25% until quoted — that band is the second-largest uncertainty after board size.

§4 — Operations Detail (the uploaded tool report had zero operations)

Operation	Machine	Rate £/hr	Cycle (s)	OEE	Machine £	Labour £	Op Total £
SMT Placement & Reflow	smt-high-speed-line	78.38	303	82%	8.06	0.55	8.60
TH Insertion & Hand Soldering	bench-assembly	0.36	24	100%	0.00	0.04	0.05
BGA X-Ray Inspection	xray-bga-inspection	47.63	475	90%	6.99	0.86	7.84
ICT / Functional Test	ict-automotive	58.39	270	95%	4.61	0.49	5.10
TOTAL					19.65	1.94	21.59

Bare board is bought in at £22.31 each (costed separately through the PCB-fab module: £18.06 board material and fab processing + tooling, packaging, logistics, overhead and margin) and enters this assembly as material, which is how a contract assembler actually buys it.

§5 — Uncertainty and Sensitivity

Driver	Basis	Effect on total
Board size / layer count	scaled from photos, not measured	±8% (±£15)
Component prices	class medians, not quoted	±25% of BOM → ±£28
BGA X-ray method	engine assumes 100% offline X-ray (475 s/board)	inline AXI at 45 s → £176.40 (–£8.36)
Assembly complexity	'high' — >300 parts, BGAs, double-sided	±10% of SMT → ±£1
Overhead / margin	9% / 8% assumed	±1pt each → ±£3

Working range £165 – £215 per board. At 150,000/yr, 100% offline X-ray would need 3.3 dedicated machines; inline AXI is the industry norm at this volume and is shown above as the realistic lower case. The headline keeps the engine's conservative default — an over-estimate is a negotiating position, an under-estimate is a signed mistake.

§6 — Gap Analysis vs the Uploaded Tool Report

#	Gap	What the tool report did	Corrected here	Severity
1	Wrong commodity	Costed as pcb_fab (bare board only), with one material line passed straight through.	A populated assembly must be costed as pcba . Bare board is one input to it, not the whole part.	Critical
2	Zero operations	Process £0.00, Labour £0.00, "Operations: 0" — no placement, reflow, inspection or test.	4 operations, £21.59 of conversion cost. A 755-placement safety ECU cannot assemble itself.	Critical
3	Volume ignored	Report shows "Annual volume: —"; tooling amortised over 5,000 parts.	150,000/yr applied. Tooling falls from £4.52-equivalent to £0.31/part.	Critical
4	No BOM	One line: mat-virtual £82.43. No component lines, so the largest cost bucket is unauditable.	20 BOM lines, £110.99, each flagged confirmed or estimated (§3).	Critical
5	Wrong-commodity advice	§11 recommended "HPDC → lost foam; forging → closed-die" and a material-utilisation lever — casting and forging advice on a PCB.	Levers gated to the commodity actually being costed.	Major
6	Carbon = 0.00	Net weight 0.000 kg → 0.00 kgCO2e for a 500 g electronics assembly.	Carbon must come from board area, copper mass and component class, not net weight.	Major
7	Confidence not enforced	Published a headline while self-reporting "Low confidence, 1 data point auditable".	A costing with zero operations should be blocked, not rendered.	Major
8	Currency	Reported in CNY with an FX note, when GBP was asked for.	Reported in GBP throughout.	Minor

§7 — Net Effect

	Total per board	Material	Process + Labour	Basis
Uploaded tool report	£98.06	£82.43	£0.00	0 operations · volume not applied · no BOM
Corrected (this report)	£184.76	£134.23	£21.59	4 operations · 150,000/yr · 20-line BOM
Difference	+£86.70	+£51.80	+£21.59	tool under-states by 47%

The gap is not a disagreement about component prices — the two BOM values are within a quarter of each other. It is that the uploaded report contained **no assembly cost whatsoever** and never applied the 150,000/yr volume. Taking £98/board into a supplier meeting for this assembly would have put the floor price roughly half where it belongs.